PLATE 01 · EXPLODED ASSEMBLY · SWEPT-WING TWIN-ENGINE TRANSPORT

ANATOMY OF AN AIRPLANE

Fourteen components pulled apart along their attachment lines — what each is called and why, what it does, the quirk worth remembering, and how it shapes the flow around the vehicle.

airplane ← Fr. aéroplane ← Gk. aēr “air” + planos “wandering” (or Fr. planer, “to glide”)

HOW TO READ THIS PLATE

Primary structure · Control & high-lift surfaces · Propulsion

Dashed lines show where each part fits back on. Every card gives its **ROOT** (etymology), then **USE**, **QUIRK** and **AERO** (its effect on the flow).

1 RADOME & COCKPIT

radome = radar + dome (1940s) · cockpit ← 1580s pit for cockfighting, later a warship’s junior-officers’ quarters

USE The nose cone houses the weather radar; behind it the flight deck holds crew, instruments and controls.

QUIRK The radome is fiberglass or composite, not aluminum — metal would block the radar beam.

AERO Sets the stagnation point where the flow splits; a smooth taper lets pressure recover gently so the boundary layer stays attached.

2 PITOT TUBE

eponym · Henri Pitot, French engineer who used one to gauge the Seine’s current in 1732

USE Senses total pressure; paired with static ports it gives airspeed from $q = p_t - p_s = \tfrac{1}{2}\rho V^2$.

QUIRK Electrically heated against ice — iced-over pitots fed false airspeeds in the 2009 Air France 447 accident.

AERO Measures the flow rather than shaping it; it is mounted where local flow is clean so the reading tracks the freestream.

3 FUSELAGE

Fr. fuselé “spindle-shaped” ← fuseau “spindle” ← Lat. fusus

USE The structural backbone: carries crew and payload and ties the wings to the tail.

QUIRK Semi-monocoque: the skin itself carries load. As a pressure vessel, every pressurization cycle counts against its fatigue life.

AERO Little lift, lots of skin-friction and form drag; a high fineness ratio (long and slender) keeps parasite drag down.

4 WING · AIRFOIL

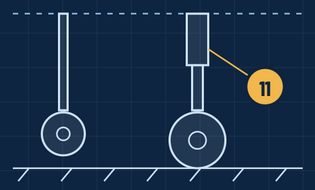
wing ← ON vængr · airfoil = air + foil ← Lat. folium “leaf”

USE Makes the lift, $L = \tfrac{1}{2}\rho V^2 S \cdot C_L$. Its internal wing box also holds most of the fuel.

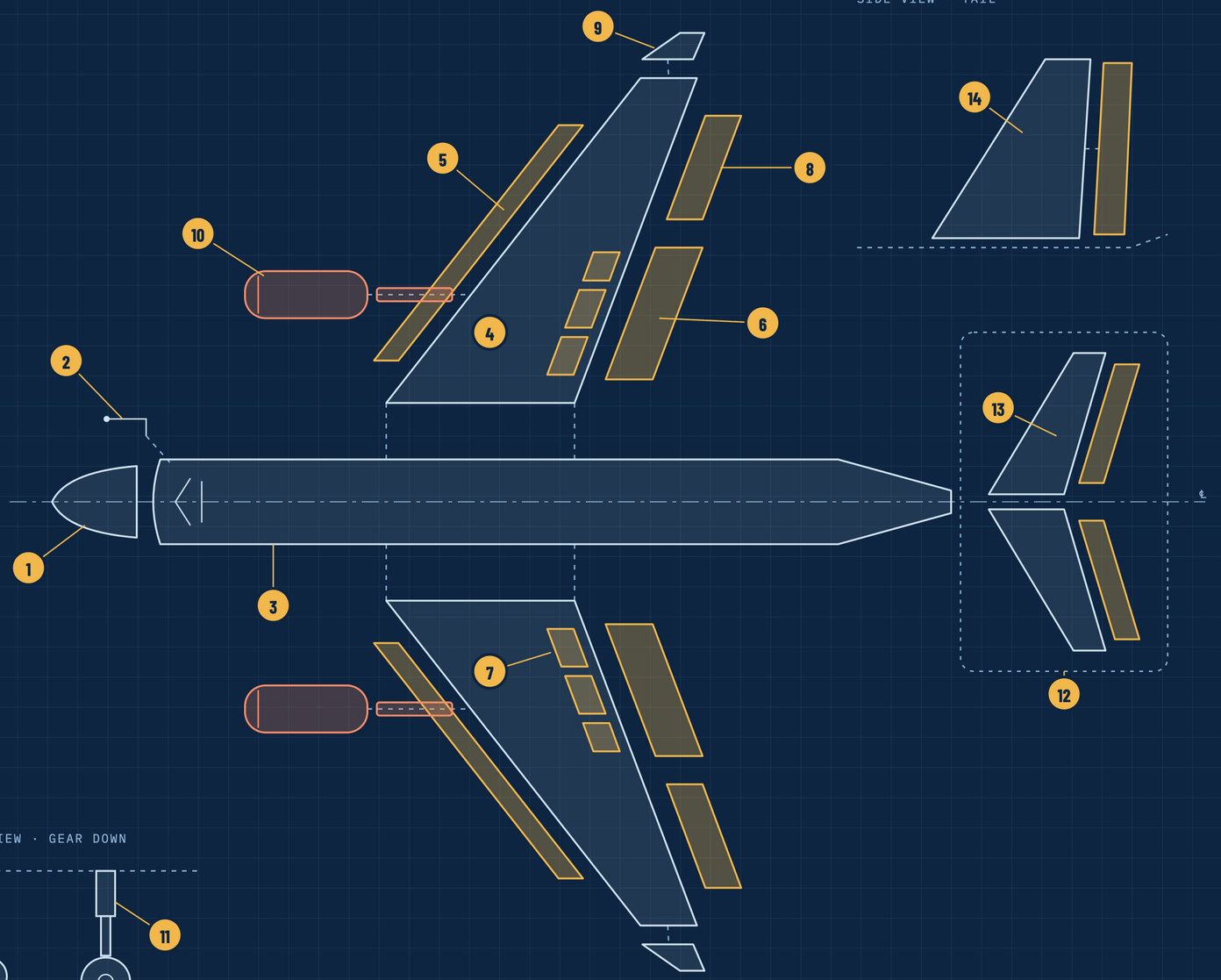
QUIRK A “wet wing” puts fuel weight right where the lift is, relieving the bending moment at the root.

AERO Camber and angle of attack set lift; sweep delays shocks near Mach 1; high aspect ratio cuts induced drag; dihedral adds roll stability.

SIDE VIEW · GEAR DOWN



NOSE ← PLAN VIEW, FROM ABOVE → TAIL



SIDE VIEW · TAIL

5 LEADING-EDGE SLATS

slat ← OFr. esclat “splinter, fragment” (modern éclat)

USE Slide forward and down off the leading edge for takeoff and landing.

QUIRK Early Handley Page automatic slats needed no actuator — suction at high angle of attack pulled them open.

AERO The slot feeds high-energy air over the upper surface, keeping flow attached and pushing the stall to a higher angle of attack.

6 FLAPS

ME flappe — imitative, the sound of a slap

USE Trailing-edge high-lift devices for slow flight: takeoff, approach and landing.

QUIRK Fowler flaps roll aft on tracks before drooping, adding area as well as camber; the canoe-shaped fairings under an airliner’s wing hide those tracks.

AERO Raise $C_{L,max}$ and lower stall speed, at the cost of drag and a nose-down pitching moment the tail must trim out.

7 SPOILERS

spoil ← OFr. espoillier ← Lat. spoliare “to strip, plunder”

USE Upper-surface panels that rise to dump lift: speedbrake in flight, roll assist, ground spoilers at touchdown.

QUIRK On landing they deploy automatically, shifting weight onto the wheels so the brakes can actually grip.

AERO Deliberately separate the flow — literally “spoiling” lift and adding drag over part of the span.

8 AILERONS

Fr. aileron “little wing” ← aile “wing” ← Lat. ala

USE Deflect in opposite directions to roll the airplane about its longitudinal axis.

QUIRK Adverse yaw: the down aileron makes more lift and more induced drag, dragging the nose away from the turn. Rudder, differential or Frise ailerons cancel it.

AERO Change local camber so lift becomes asymmetric across the span.

9 WINGLET

wing + -let (diminutive) · developed by NASA’s Richard Whitcomb in the 1970s

USE An upturned tip that recovers energy otherwise lost to the wingtip vortex.

QUIRK Pays off most on long cruise legs; on short hops its weight and extra wing bending can eat the gain.

AERO High-pressure air under the wing curls around the tip to the low-pressure top; the winglet weakens that vortex, cutting induced drag much like extra span.

10 NACELLE & PYLON

nacelle ← Fr. “small boat” ← Lat. navicella ← navis “ship” · pylon ← Gk. pylōn “gateway”

USE The nacelle fairs the engine; the pylon hangs it ahead of and below the wing.

QUIRK In a high-bypass turbofan most thrust comes from the fan, not the core. Engine mass hung forward also helps damp wing flutter.

AERO Supplies thrust but adds interference drag at the wing–pylon joint; small nacelle strakes shed a vortex that keeps flow on the wing at high angle of attack.

11 LANDING GEAR

a.k.a. undercarriage · gear ← ON gervi “equipment” · oleo strut ← Lat. oleum “oil”

USE Carries the airplane on the ground, absorbs touchdown, steers (nose gear) and brakes (mains).

QUIRK Oleo-pneumatic struts push oil through an orifice against compressed nitrogen: a spring and a damper in one tube.

AERO Extended, it is a major parasite-drag source — why fast airplanes retract it, and why lowering it is part of slowing down on approach.

12 EMPENNAGE

Fr. empenner “to feather an arrow” ← penne “feather” ← Lat. penna

USE The whole tail group: horizontal and vertical stabilizers plus their control surfaces (dashed outline, 12).

QUIRK Named for arrow fletching — and it works the same way, putting area far behind the center of gravity.

AERO On its long moment arm, any disturbance creates a restoring moment that weathervanes the nose back to the relative wind: static stability in pitch and yaw.

13 H. STABILIZER & ELEVATOR

stabilis “standing firm” (Lat.) · elevator ← Lat. elevare “to lift up” ← levis “light”

USE The stabilizer gives pitch stability; the elevator controls pitch about the lateral axis.

QUIRK On most airliners the tail pushes down to balance a nose-heavy layout, and the whole stabilizer moves to trim.

AERO That down-load means the wing must lift slightly more than the airplane weighs — a penalty called trim drag.

14 V. STABILIZER & RUDDER

fin ← OE finn · rudder ← OE rōþor “paddle, oar”

USE The fin gives directional stability; the rudder controls yaw about the vertical axis.

QUIRK The rudder doesn’t turn the airplane — banking does. It coordinates turns, fights crosswinds, and is sized to hold an engine-out.

AERO A symmetric airfoil set on edge: sideslip gives it an angle of attack, and its side force swings the nose back into the wind.

THREE AXES

ALL THROUGH THE CENTER OF GRAVITY

ROLL
Longitudinal axis, nose to tail · **ailerons (8)**
← OFr. roller ← Lat. rotula “little wheel”

PITCH
Lateral axis, wingtip to wingtip · **elevator (13)**
← ME picchen “to thrust, set”; a ship’s bow pitching

YAW
Vertical axis, top to bottom · **rudder (14)**
nautical, 1500s; origin unknown

FOUR FORCES

STEADY LEVEL FLIGHT: $L = W$, $T = D$

LIFT
$L = \tfrac{1}{2}\rho V^2 S \cdot C_L$
← ON lypta, from lopt “air, sky”

WEIGHT
$W = mg$
← OE gewiht

THRUST
$T \approx \dot{m}(V_e - V_0)$
← ON þrýsta “to press”

DRAG
$D = \tfrac{1}{2}\rho V^2 S(C_{D0} + C_L^2/\pi eAR)$
← OE dragan “to draw”